

The radial extension is 3-Lipschitz

MathlibAnnex.Sphere.lipschitzWith_radialExtension

theorem

Separates radial variation from the change of unit direction to obtain the global bound with coefficient $1 + 2$.

Statement

For real normed spaces X, Y and a bijective chord-metric isometry $\Delta : S_X \rightarrow S_Y$, define $R_\Delta(0) = 0$ and $R_\Delta(x) = \|x\|\Delta(x/\|x\|)$ for $x \neq 0$. Here $S_X = \{u \in X : \|u\| = 1\}$ and $S_Y = \{v \in Y : \|v\| = 1\}$ are the unit spheres. Then for every $x, y \in X$,

$$\|R_\Delta(x) - R_\Delta(y)\| \leq 3\|x - y\|.$$

Assumptions

The sphere distances are inherited from the ambient norms. The map Δ is a bijective isometry, so it preserves these chord distances and sends unit vectors to unit vectors. No completeness, finite-dimensionality or nontriviality assumption is made.

Conclusion

The forward radial map is globally 3-Lipschitz. Applying the same theorem to Δ^{-1} gives the separately stated 3-Lipschitz bound for $R_{\Delta^{-1}}$. Together with the inverse identities, this makes the radial bijection a homeomorphism with Lipschitz inverse.

The constant 3 is a proved upper bound. This statement makes no optimality assertion and supplies no ambient distance-preserving or linear-extension conclusion. Both directions of the homeomorphism use the same radial maps as in the definitions.

Proof route

First assume $0 < \|y\| \leq \|x\|$, and put $\hat{x} = x/\|x\|$, $\hat{y} = y/\|y\|$. Split the difference at the smaller radius:

$$\begin{aligned} R_\Delta(x) - R_\Delta(y) &= (\|x\| - \|y\|)\Delta(\hat{x}) \\ &\quad + \|y\|(\Delta(\hat{x}) - \Delta(\hat{y})). \end{aligned}$$

The first direction has norm one and Δ preserves chord distances. Hence

$$\begin{aligned} \|R_\Delta(x) - R_\Delta(y)\| &\leq |\|x\| - \|y\|| + \|y\|\|\hat{x} - \hat{y}\| \\ &\leq \|x - y\| + 2\|x - y\| = 3\|x - y\|. \end{aligned}$$

The first bound is the reverse triangle inequality. The second uses the angular lemma with its three hypotheses $x \neq 0$, $y \neq 0$ and $\|y\| \leq \|x\|$. The following calculation gives its coefficient 2.

▼ Proof steps

1. Set $r = \|y\|/\|x\|$. The assumed order and positivity imply $0 \leq r \leq 1$ and $r\|x\| = \|y\|$. Scaling the normalized difference gives

$$\|y\|(\hat{x} - \hat{y}) = rx - y = r(x - y) + (r - 1)y.$$

$$\text{Consequently } \|y\|\|\hat{x} - \hat{y}\| = \|rx - y\|.$$

2. The first summand satisfies $\|r(x - y)\| = r\|x - y\| \leq \|x - y\|$. For the second,

$$\begin{aligned} \|(r - 1)y\| &= (1 - r)\|y\| \\ &\leq (1 - r)\|x\| = \|x\| - \|y\| \\ &\leq \|x - y\|. \end{aligned}$$

The triangle inequality now yields $\|rx - y\| \leq 2\|x - y\|$, which is the required angular bound.

3. Insert this angular bound into the displayed radial decomposition. If the norm order is reversed, interchange x, y and use the symmetry of $\|x - y\|$ and of the image distance. If one vector is zero, norm preservation gives the sharper equality $\|R_\Delta(x) - R_\Delta(0)\| = \|x\|$, so the bound by $3\|x\|$ follows; the two-zero case is immediate.

Main citations

- Exact declaration and its source — `MathlibAnnex.Sphere.lipschitzWith_radialExtension`
- The radial extension — `MathlibAnnex.Sphere.radialExtension`
- The angular estimate at the smaller radius — `MathlibAnnex.Sphere.scaled_normalize_dist_le_two`
- The radial and angular decomposition in the proof — `MathlibAnnex.Sphere.radialExtension_dist_le_three_of_norm_le`
- The extension at zero — `MathlibAnnex.Sphere.radialExtension_zero`
- The nonzero defining formula — `MathlibAnnex.Sphere.radialExtension_of_ne_zero`
- Norm preservation — `MathlibAnnex.Sphere.radialExtension_norm`
- The inverse Lipschitz estimate — `MathlibAnnex.Sphere.lipschitzWith_radialExtension_symm`
- Inverse identities — `MathlibAnnex.Sphere.radialExtension_rightInverse`
- The radial homeomorphism — `MathlibAnnex.Sphere.radialExtensionHomeomorph`

Lean source signature (exact)

```
theorem lipschitzWith_radialExtension
  (e : Metric.sphere (0 : X) 1 ≈i Metric.sphere (0 : Y) 1) :
  LipschitzWith 3 (radialExtension e)
```

In the source	Mathematical meaning
$(e : \text{Metric.sphere } (0 : X) \ 1 \approx_i \text{Metric.sphere } (0 : Y) \ 1)$	The same bijective chord-distance isometry $\Delta : S_X \rightarrow S_Y$, where $S_X = \{x : \ x\ _X = 1\}$ and $S_Y = \{y : \ y\ _Y = 1\}$. The surrounding binders make X, Y real normed spaces.
In the source	Mathematical meaning
<code>radialExtension e</code>	The map $R_\Delta : X \rightarrow Y$, with $R_\Delta(0) = 0$ and $R_\Delta(x) = \ x\ _X \Delta(x/\ x\ _X)$ for $x \neq 0$.
<code>LipschitzWith 3 (radialExtension e)</code>	For every $x, y \in X$, $\ R_\Delta(x) - R_\Delta(y)\ _Y \leq 3\ x - y\ _X$. The nonnegative constant 3 is the proved bound, not an extra input or an optimality claim.

▼ Exact surrounding binder context (separate excerpt)

```
namespace MathlibAnnex
namespace Sphere

universe u v

variable {X : Type u} {Y : Type v}
  [NormedAddCommGroup X] [NormedSpace ℝ X]
  [NormedAddCommGroup Y] [NormedSpace ℝ Y]
```

▼ Exact content identity

Declaration: `MathlibAnnex.Sphere.lipschitzWith_radialExtension`

Accepted content SHA-256: 01fbb3907373edf7dcdd7cb95a51fbeb2b5c73335d05a1def4972ee8b752871b

Accepted source guide SHA-256: 87ed83f724d62dfbd967187494fe9beedd18d6a8a0bd678b0f56f3486cf1130a

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73